Program: Artificial Intelligent Program.
Level : 2nd Year
Course Code: IS200
Course Name: Database Systems
No. of Questions : 3

Zagazig National University

Mid Term Exam
Term No.: First-term
Date : 16/11/2024
Time : 1 hour
No. of Page : 2 pages
Full Mark : 15

Student Name: Student Code: .....

Please read the following questions carefully and answer all the following questions

Q1: Choose the correct answer (5 Points)

1. The total participation by entities is represented in E-R diagram as
a) Dashed line b) Double rectangle c) Double line d) Circle
2. A _____ attribute need not be physically stored within the database
a).Single valued b). Derived c). Multi valued d). Composite
3. .....is the capacity to change the internal schema without having to change the conceptual schema and the external schemas need not be changed as well.
a) Logical data independence b) Physical data independence
c)Logical data independence d) Physical data dependence
4. ..... Stored by the DBMS in the form of a database catalog or dictionary
a) Data b) meta-data c) database d) information
5. .....end user, access database occasionally when needed
a) Casual b) naive c) sophisticated d) Stand-alone
6. Which of the following can be a multivalued attribute?
a) Name b) Date_of_birth c) Phone_number d) All of the mentioned
7. Database _____ which is the logical design of the database, and the database _____ which is a snapshot of the data in the database at a given instant in time.
a) Instance, Schema b) Relation, Schema c) Relation, Domain d) Schema, Instance
8. ..... Of the schema , Describes physical storage structure of the database.
a) Internal level b) Conceptual level c) External level d)All of the above
9. A _____ is an association among several entities.
a) Relationship b) Association c)Set d) Combination
10. A ..... represents the number of entities to which another entity can be associated
a). cardinality ratio b). table c). schema d). information

Q2: True / false Question (5 Points)

1. For a weak entity to be meaningful, it must be associated with another entity, called the Owner entity T
2. A person is an example of an attribute F
3. SQL stands for Standard Query Language. T
4. ERDs illustrate the logical structures of DBMS. T
5. A database is called "self-describing" because it contains a description of itself. T
6. In a database, data is stored in spreadsheets which have rows and columns. F
7. An attribute is known as a row in most databases T
8. Software packages, called database management systems, can do the job of manipulating data for you. T
9. A DBA can assign passwords to prevent unauthorized users from accessing the data F T
10. A query is a question represented in a way that the DBMS can recognize and process T

الاسم: ..... رقم السكشن: .....

Q1) Consider the following set of requirements for a UNIVERSITY database: **(7 Points)**

- Professors have an SSN, a name, an age, a rank, and a research specialty.
- Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget.
- Graduate students have an SSN, a name, an age, and a degree program (e.g., M.Sc. or Ph.D.).
- Departments have a department number, a department name, and a main office.
- Each project is managed by one professor (known as the project's principal investigator).
- Each project is worked on by one or more professors (known as the project's co- investigators). Professors can manage and/or work on multiple projects.
- Each project is worked on by one or more graduate students (known as the project's research assistants). Graduate students can work on multiple projects.
- Professor's work in one or more departments and for each department that they work in, a time percentage is associated with their job.
- Graduate students have one major department in which they are working on their degree. • Each graduate student has another, more senior graduate student (known as a student advisor) who advises him or her on what courses to take.

Design an ER Diagram to and specify key attributes of each entity type and structural constraints on each relationship type.

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